

Smart Playlist Generation Using Clustering and Spotify Audio Features

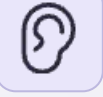
A data-driven approach to automated playlist creation

Dataset: 5,000 Spotify tracks





Project Objective

-  **Automatically generate playlists grouped by musical similarity**
-  **Ensure each playlist has between 50 and 250 songs**
-  **Rely purely on audio features, not metadata**
-  **Prioritize musical traits that align with human listening perception**

Our Approach

- 1 Collect audio features from Spotify
- 2 API Normalize features using
- 3 MinMaxScaler Apply custom feature weights to emphasize perceptual relevance
- 4 Perform initial clustering using
- 5 KMeans Re-cluster with KMeansConstrained to enforce playlist size
- 6 constraints Generate final playlists from resulting clusters

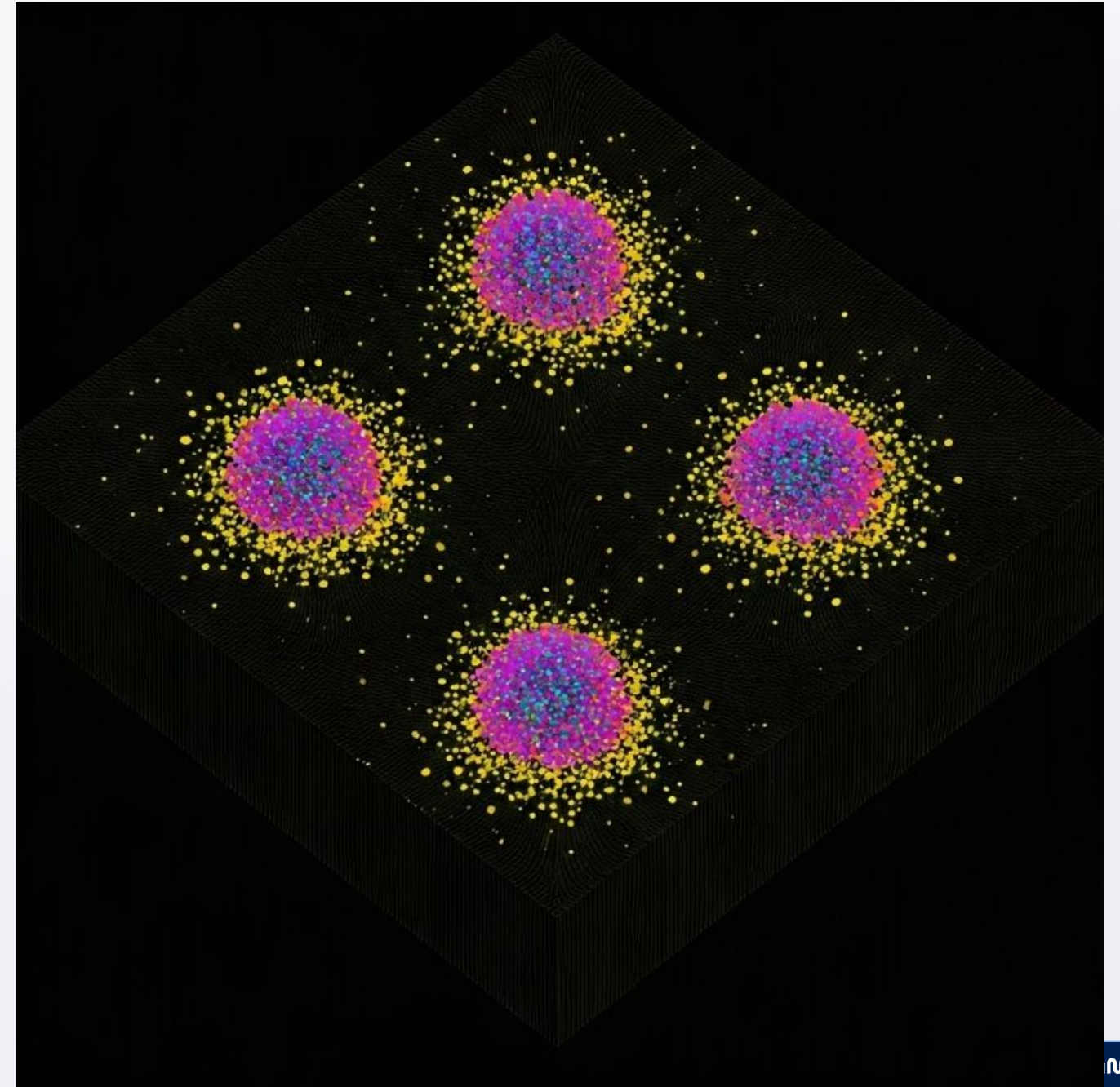
Initial Clustering with

KMeans

Chose $k=5$ based on silhouette and inertia scores then sub-clustering

Good cluster separation, but...

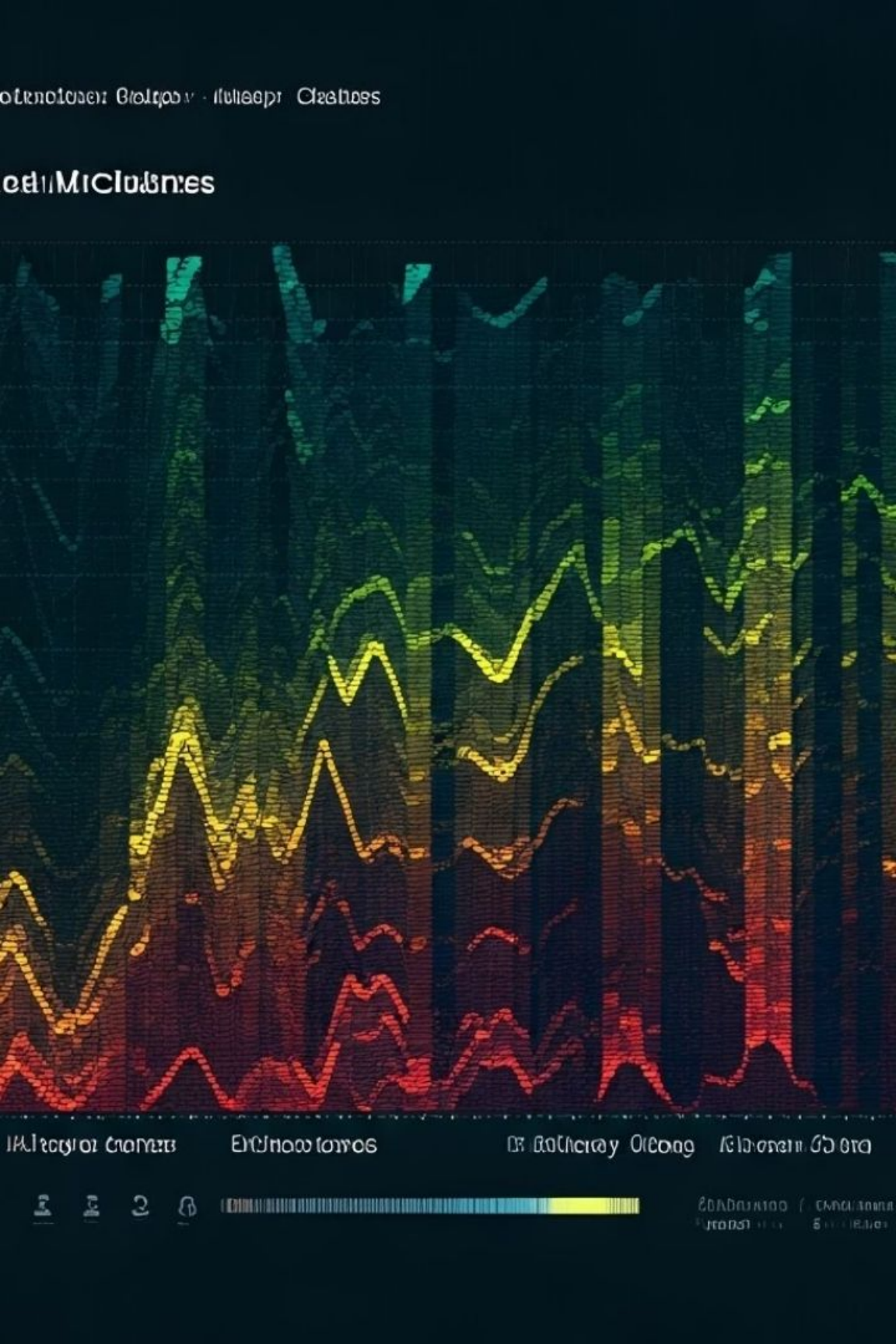
Playlist sizes became inconsistent when sub-clustered



The Problem



Some clusters had <50 songs, others >250



Solution: KMeansConstrained + Feature Weighting

Size Constraints

Supports size_min and size_max parameters

Weighted Features

Re-applied clustering with weighted features to reflect listening experience

Perceptual Emphasis

- Acousticness & Instrumentalness ($\times 1.5$): distinguish natural vs. electronic and vocal vs. instrumental music
- Danceability ($\times 1.2$): emphasize rhythm
- Duration, Loudness, Speechiness ($\times 0.5\text{--}0.8$): reduced impact due to lower perceptual relevance

Result

45 balanced, perceptually meaningful playlists

Features Used

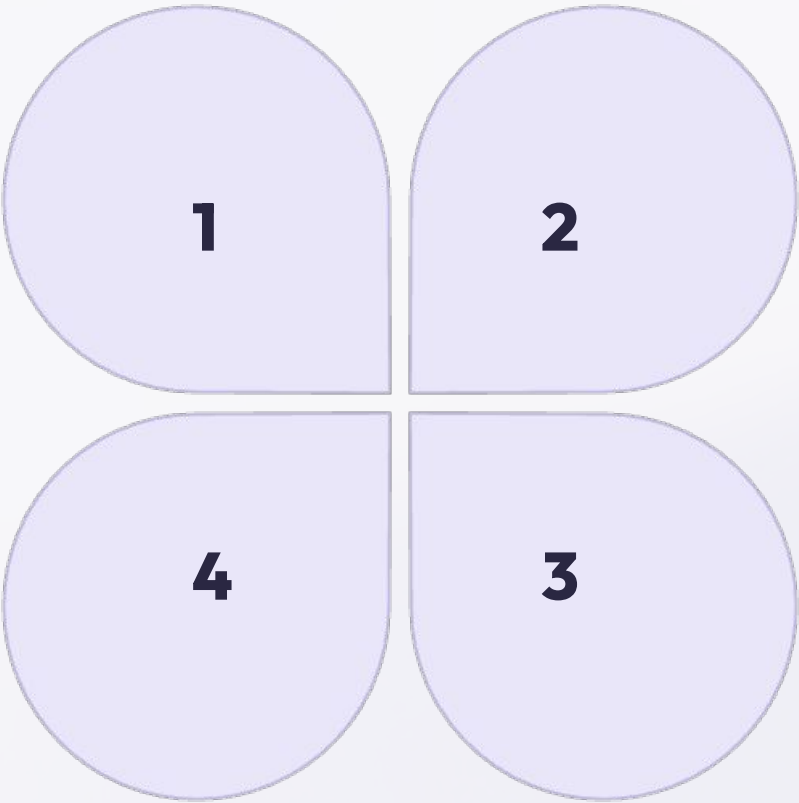
Weighted Features

(High)

- acousticness (1.5)
- instrumentalness (1.5)
- danceability (1.2)

Dropped Features

- key, mode, time_signature (low variance)
- artist, genre, song_name (non-numeric)



Weighted Features

(Medium)

- energy (1.0)
- valence (1.0)
- tempo (1.0)

Weighted Features

(Low)

- speechiness (0.8)
- loudness (0.8)
- liveness (0.7)
- duration_ms (0.5)

Playlist Examples



Playlist 1 (Cluster

22) High danceability, mid-tempo, high valence

Feels like a summer chill playlist



Playlist 2 (Cluster

14) Low energy, high acousticness

Ideal for studying or relaxation



Are Spotify Features Human-Meaningful?



Mood Matching

Features like energy, tempo, valence often match human-perceived mood



Style Recognition

Acousticness and instrumentality help surface stylistic similarities



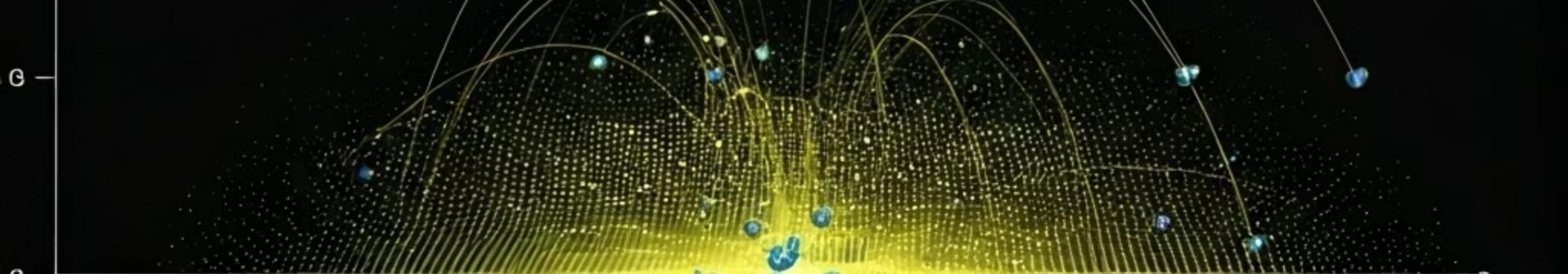
Missing Context

However, no awareness of lyrics, language, genre or cultural context



Compensation

Weighting features helps compensate for these gaps



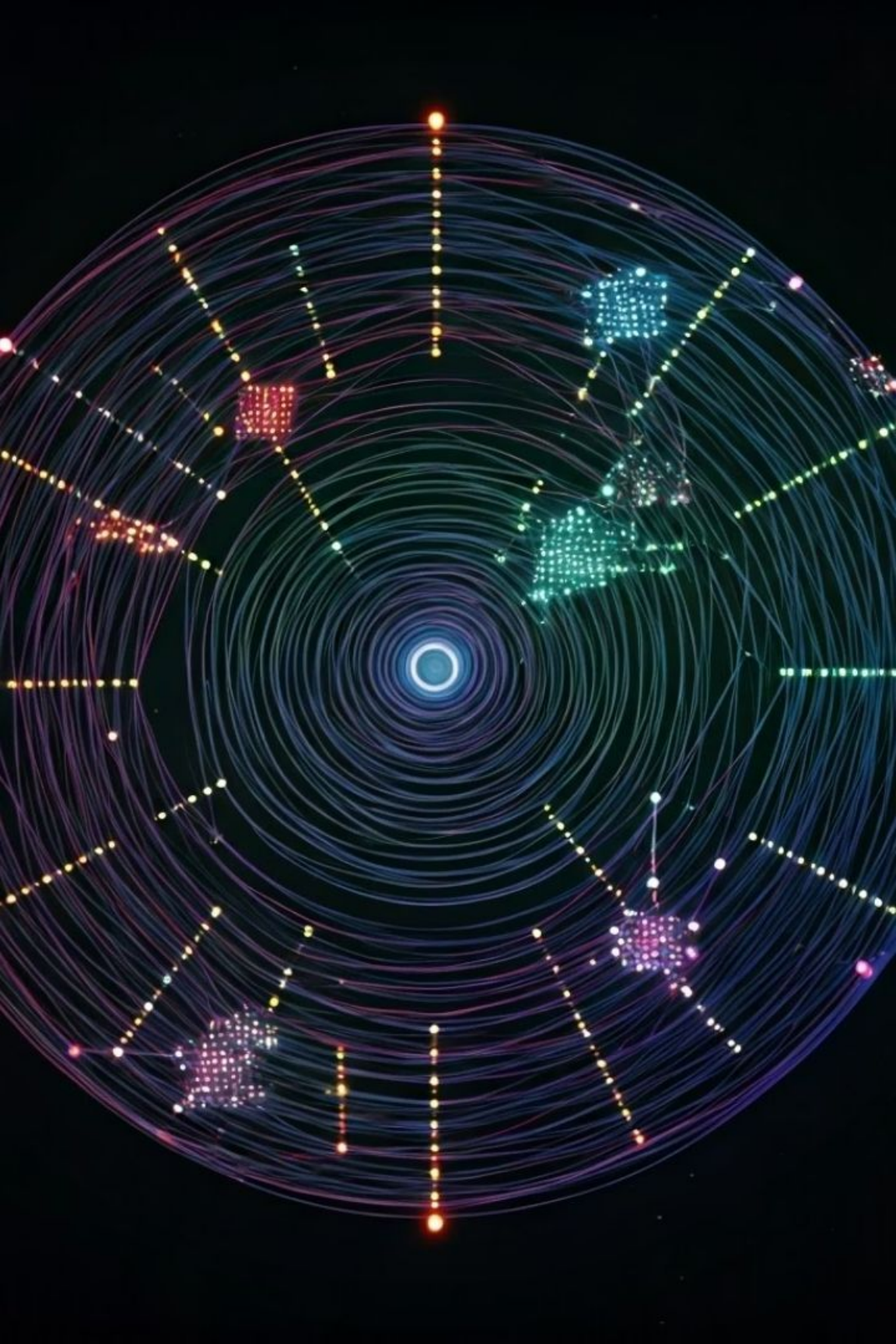
Is KMeans a Good

Fit?

Pros	Cons
Fast and easy to implement	Assumes equal-sized, spherical clusters
Interpretable and tunable	No soft membership (e.g., songs in multiple playlists)
Enhanced with feature weighting and size constraints	

Verdict:

With weighted features and constrained sizing, KMeans becomes a practical and human-aligned solution



Next Steps

Use UMAP/t-SNE for visualizing clusters

Incorporate genre tags, lyrics, or user preferences

Explore soft clustering (e.g., GMM)



Final Thought



Beyond Data

Clustering aren't just about clustering data



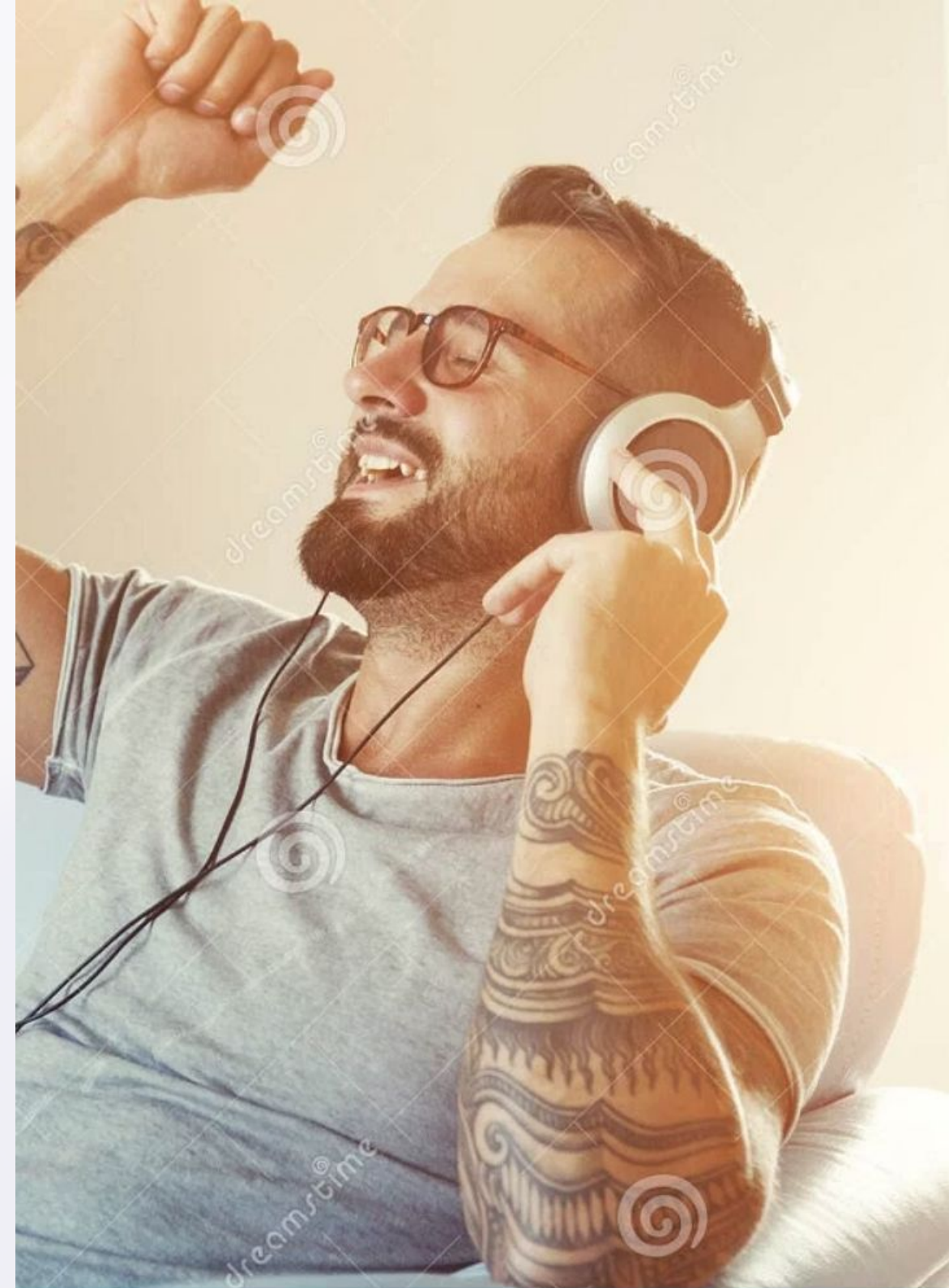
Human Perception

They're about mimicking how humans perceive, group, and enjoy music

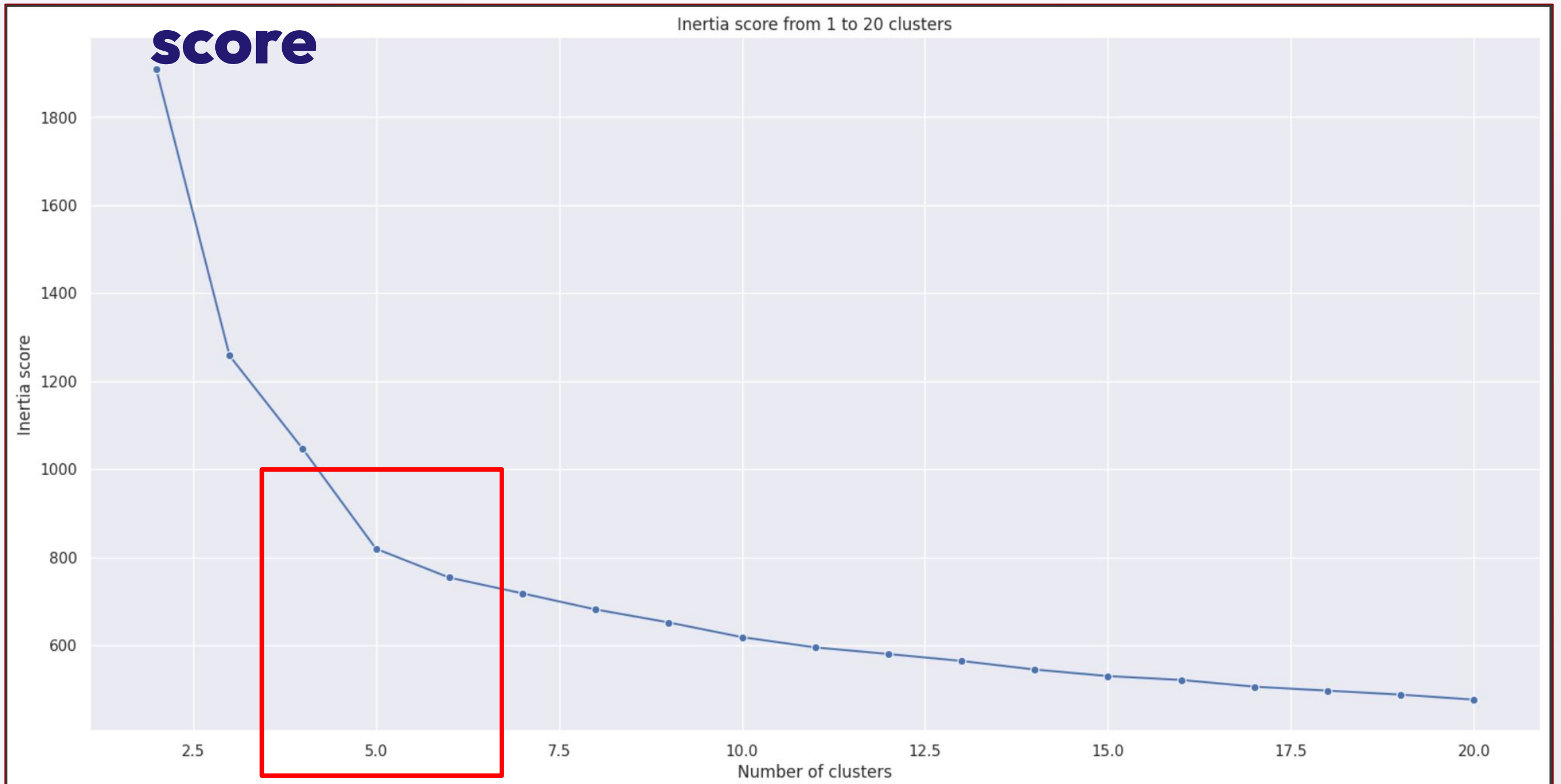


Human-Like

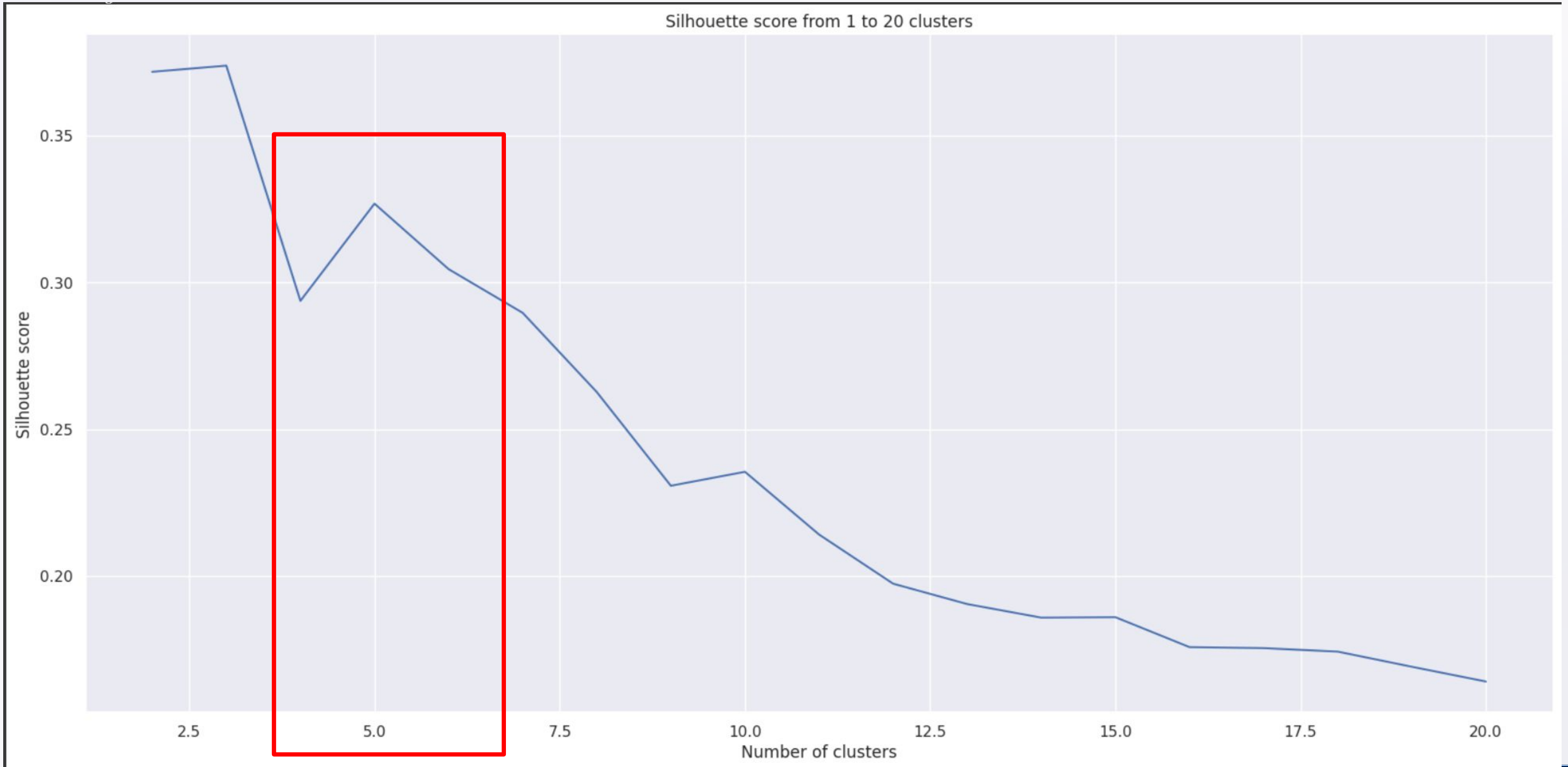
Listening features helped us teach the algorithm to "listen like a human"



First step of clustering: using Inertia

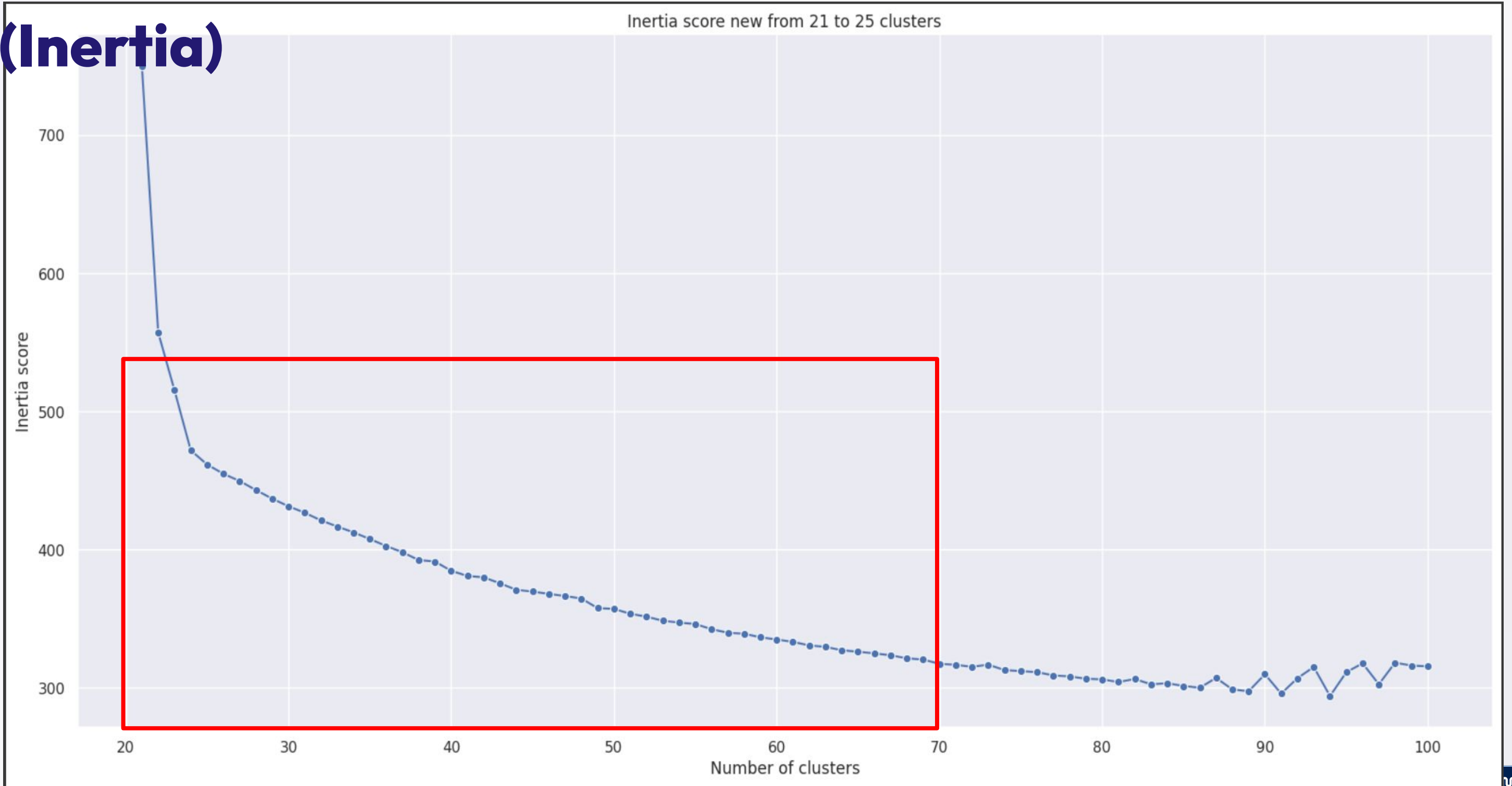


First step of clustering: using Silhouette score

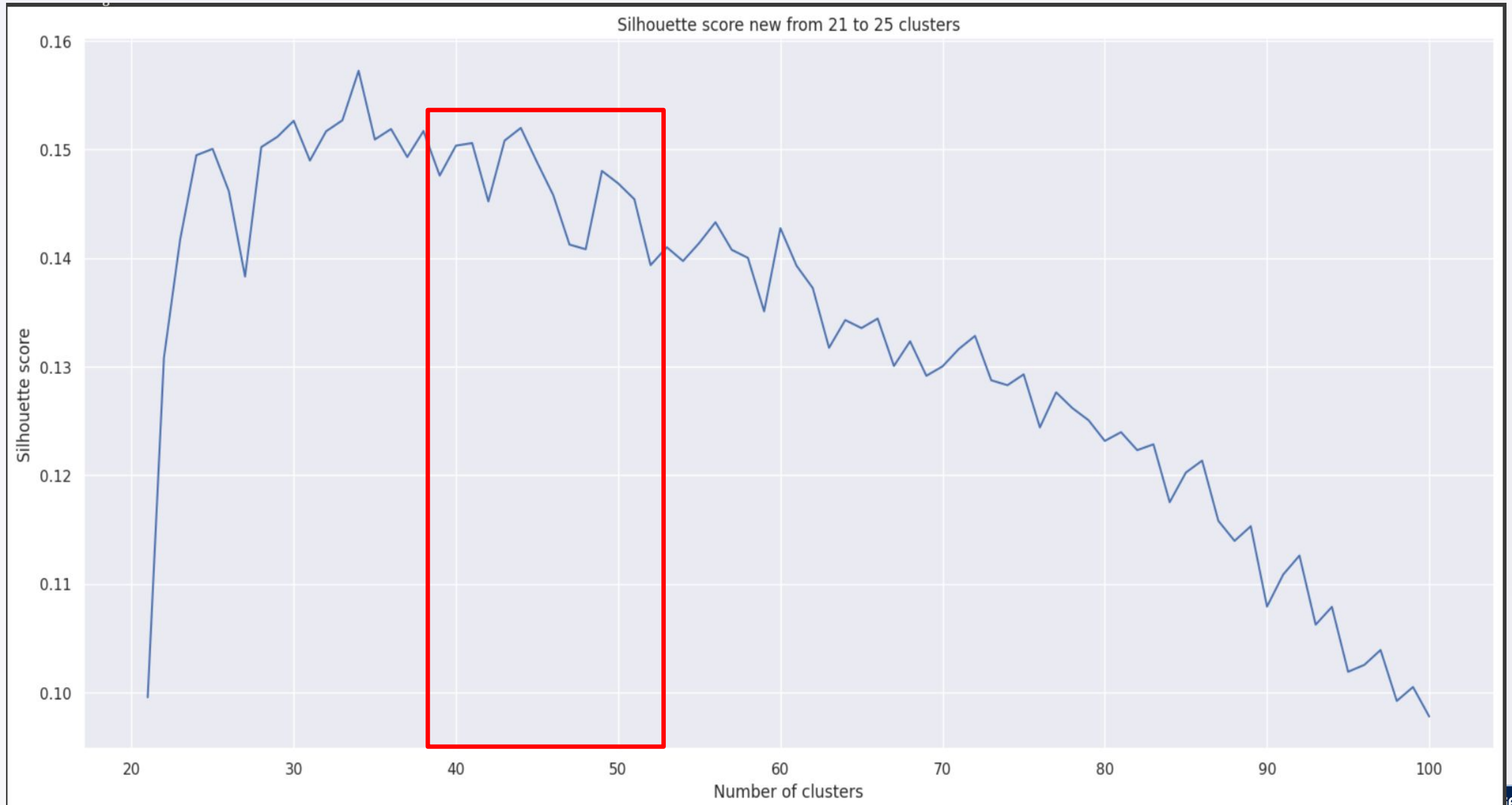


Finding n of clusters for K-Means constrained

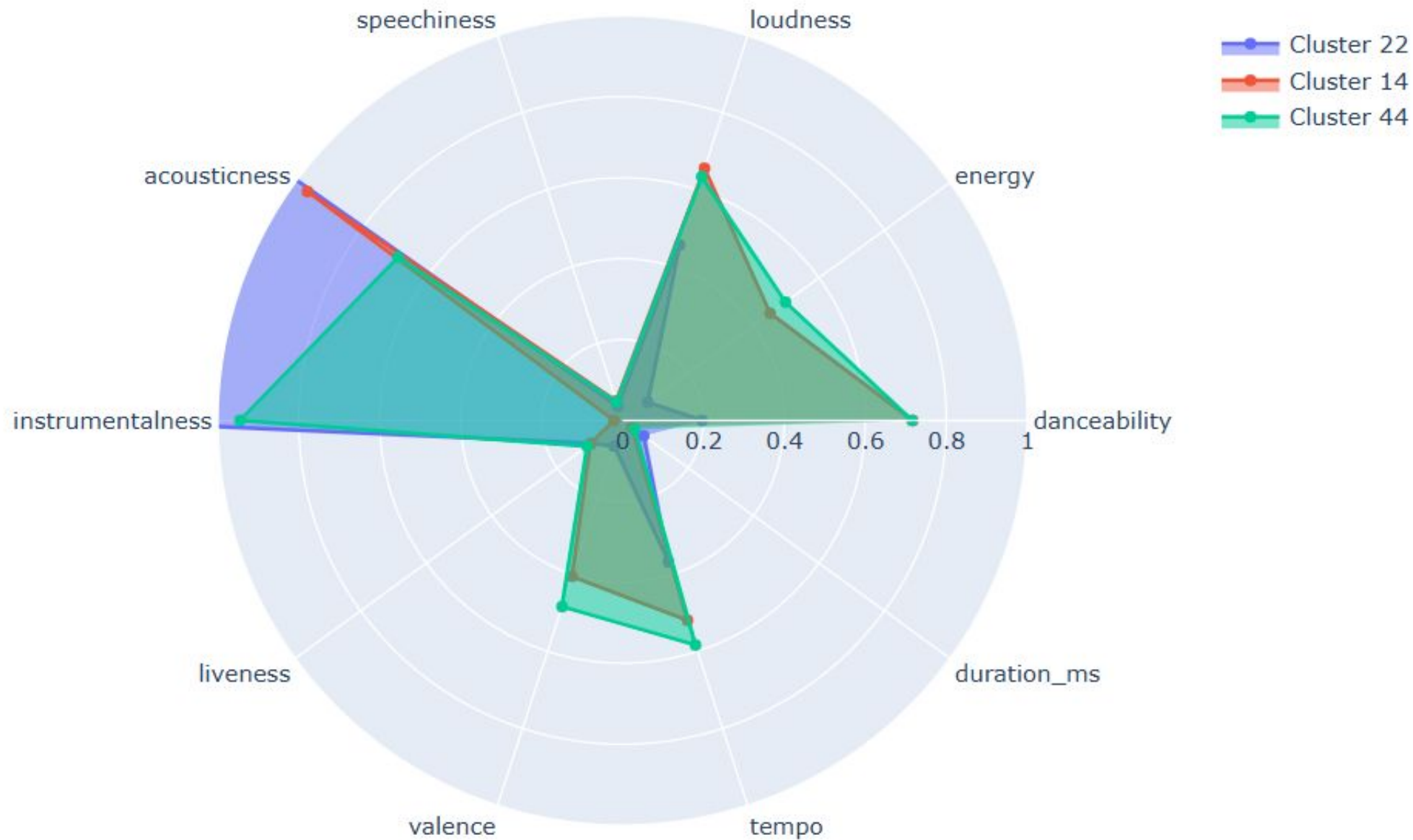
(Inertia)

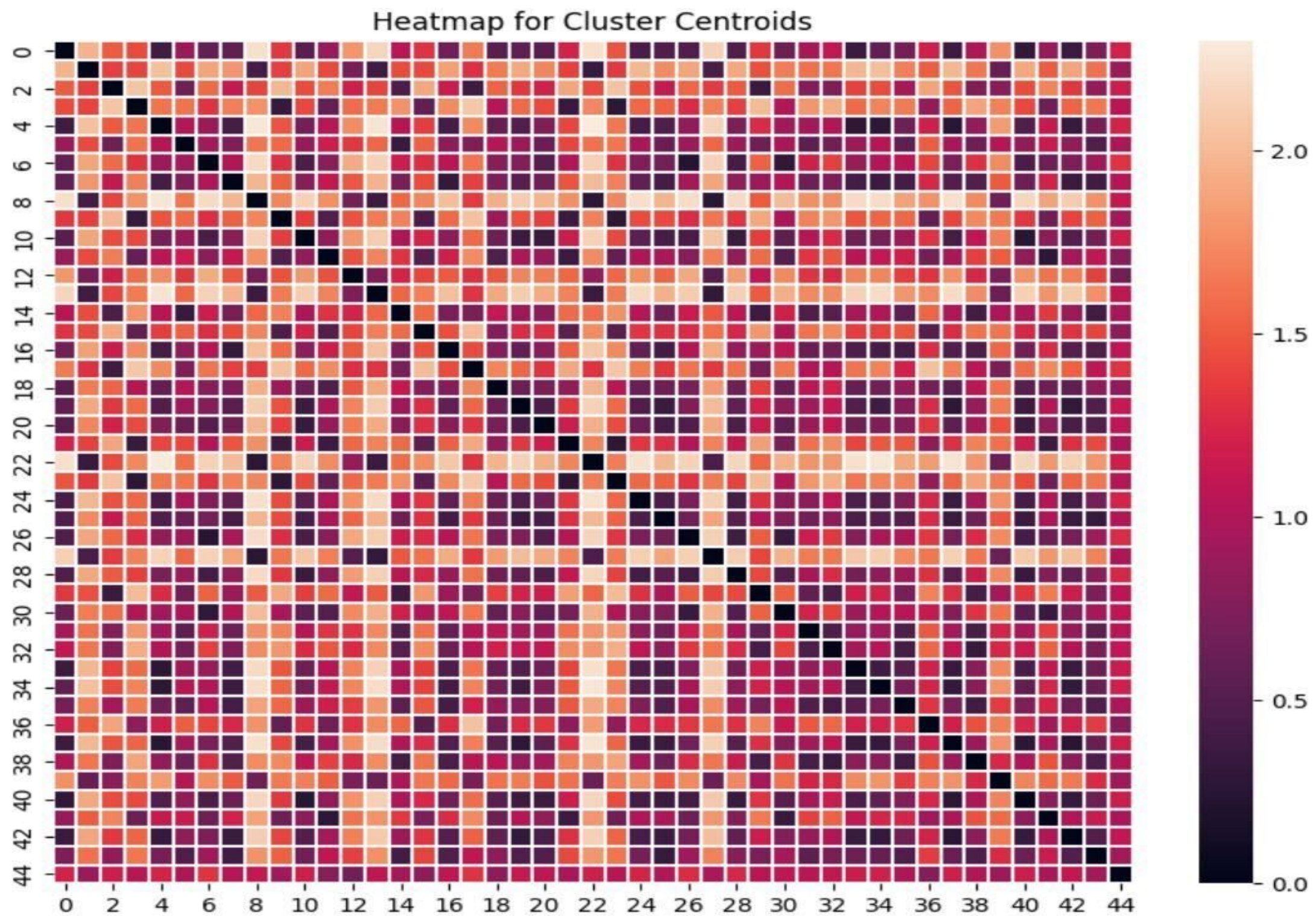


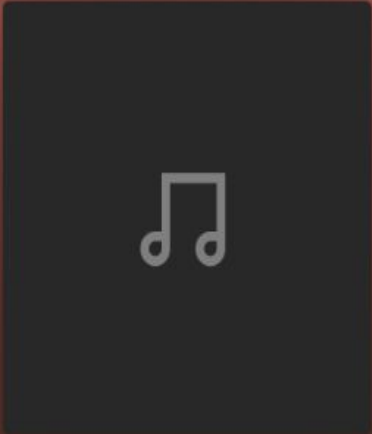
Finding n of clusters for K-Means constrained (Silhouette)



Radar chart of mean songs features by cluster







Public Playlist

my_playlist_cluster_1

this is a test list

Hassan syrian • 5 songs, 23 min 45 sec

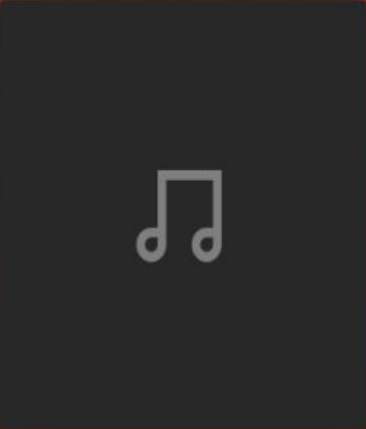


List

#	Title	Album	Date added	
1	 Serenade To A Soul Sister - Remastered Horace Silver	Serenade To A Soul Sister	26 minutes ago	6:19
2	 Warm Shadow Fink	Perfect Darkness	26 minutes ago	5:46
3	 Karoliina Kaunis Maria Gasolina	Mä Olen Sun	26 minutes ago	4:11
4	 Son of a Preacher Man Aretha Franklin	This Girl's in Love with You	26 minutes ago	3:19
5	 Beauty and the Beast - from the Soundtrack "Beauty and the Beast" Céline Dion, Peabo Bryson	Celine Dion	26 minutes ago	4:09

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Public Playlist

my_playlist_cluster_2

this is a test list

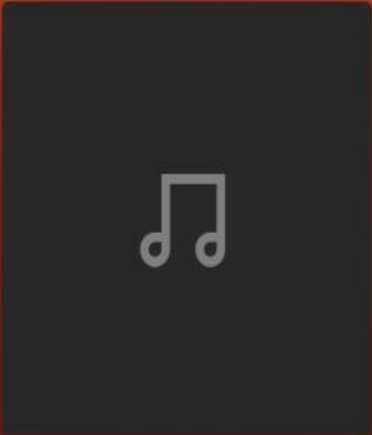
Hassan syrian • 5 songs, 33 min 55 sec

List

#	Title	Album	Date added	
1	Yanagiwara Hirotaka Shirotsubaki	Hyogo	27 minutes ago	7:27
2	unother Terekke	Improvisational Loops	27 minutes ago	1:02
3	Mussorgsky, Ravel: Pictures at an Exhibition, M. A 24: IV. Bydlo Modest Mussorgsky, Oslo Philharmonic Orchestra, Mariss Jansons	Mussorgsky & Rimsky-Korsakov: Orchestral Works	27 minutes ago	2:49
4	Choros No. 8 Heitor Villa-Lobos, Linda Bustani, Ilan Rechtman, Orquestra Sinfônica do Estado de S...	Villa-Lobos, H.: Choros, Vol. 2 - Choros Nos. 1, 4, 6, 8, 9	27 minutes ago	20:17
5	Olancha Farewell Harold Budd	Lovely Thunder	27 minutes ago	2:18

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Public Playlist

my_playlist_cluster_3

this is a test list

Hassan syrian • 5 songs, 27 min 17 sec

List

#	Title	Album	Date added	
1	 Be - Instrumental Steve Angello, Laidback Luke	Be	27 minutes ago	6:49
2	 Sonnentanz Klangkarussell	Netzwerk	27 minutes ago	6:00
3	 I Love You Always Forever Donna Lewis	Now in a Minute	27 minutes ago	3:59
4	 How to Disappear Completely Radiohead	Kid A	27 minutes ago	5:56
5	 Blue Bossa - Remastered Joe Henderson	Blue Note Trip 2: Sunset/Sunrise	27 minutes ago	4:31

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